Expansion protocol (DRAFT, not locked)
What makes an execution-time CBF-QP filter costly for frozen MAPPO formation policies?

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2 November 2026. Nothing in this document has been run. It becomes binding only when committed with advisor approval recorded.

0. Status and rules

- Study 1 (final test of 25–26 October 2026, protocol commit `9baf6d7`) stays exactly as reported; its verdict is not revisited.
- The studies below use only the reserved layout block 5,000,000–5,999,999, unused so far. Development work (gain selection, debugging) uses the validation layouts $(2{,}000{,}000 + K)$ only.
- Before any trial: code for each variant is written and unit-tested, the advisor approves margins, variants, decision rules and seeds, and the protocol is committed. Findings from Study 1 that informed this design are listed in Section 7.

1. Questions

Q1. Which design elements of the filter account for its cost in completion and formation, with the same frozen actors and matched layouts? (Study 2)
Q2. Does the cost shrink for a policy family that passes a competence gate and was trained on the test densities? (Study 3)
Q3. How do limited communication range, message dropouts and delays affect collisions and completion, and when does the guarantee stop applying? (Study 4)

2. Study 2: filter-design comparison (frozen Study-1 actors)

Arms (each on the same layouts, filter off as reference):

Arm	Change from Study 1 filter	Purpose
V0	none (two-sided margins, $\ell = 5\,\text{cm}$, equal split, no fallback)	replication on new layouts
V1	tight one-sided state-only margin (lower bound on $\ddot{h}$ only)	margin conservatism
V2	zero-command-admissible rows (δ evaluated at the held command, $\propto \lvert\omega\rvert$)	stopping admissible at $h \geq 0$
V3	V0 + brake fallback on infeasibility + nonprogress-recovery rule (fixed below)	missing fallback / recovery
V4	V0 with $\ell = 2\,\text{cm}$ in the filter only	cover-disc geometry
V5	V1 + V3	combined
B	B-match on V0 rows	brake comparator

Recovery rule (V3, V5), fixed in advance: when the centroid-to-goal distance has fallen by less than 0.05 m over 30 periods and some barrier row is active, each robot with an active obstacle row adds a tangential command component of fixed magnitude for 10 periods in the direction that increases $\lVert p - c \rVert$ fastest among the two tangents,

then resumes. (Advisors: approve or replace.)
Gain: re-selected per arm on validation layouts from $\{2, 5, 10\}\,\mathrm{s}^{-1}$ by the Study-1 rule; V2 requires its own proof check (Section 6).
Layouts: $5{,}000{,}000 + 1{,}000p + K$, $p = 0, \ldots, 8$, $K = 1, 2, 3$; 1,000 per actor and density; all arms on identical layouts.
Confirmatory criteria: per arm and density, (a)–(c) of Study 1 against filter off, same margins (unless the advisor changes them now). Family: 6 filter arms $\times$ 3 densities = 18 usefulness claims; each component one-sided at 0.05/18 (Bonferroni over claims, intersection–union within a claim). Newcombe method 10 with the continuity correction, correctly labelled.
Secondary (pre-declared, descriptive): each arm minus V0 for GOAL, collision-or-infeasible, TIMEOUT and sustained nonprogress, with 95% intervals; lost/gained transitions; infeasible states' conflicting row sets.

3. Study 3: competent policy family

Training: same MAPPO recipe, curriculum whose final stage trains on $K = 1, 2, 3$ (plus 5, 8 if the advisor wishes), 5 training seeds, budget fixed in advance. Filter never used in training (a filter-aware variant is optional, see Section 8).
Competence gate (fixed before training): pooled filter-off GOAL $\geq 60\%$ over $K = 1, 2, 3$ on validation layouts for the final checkpoint of each seed; seeds failing the gate are reported, not replaced. If fewer than 3 seeds pass, Study 3 is reported as not completed.
Actors: early, middle and late checkpoints by the Study-1 iteration rule.
Arms: off, V0, and the single Study-2 arm with the highest pooled filter-on GOAL count among arms that met (a) at all densities (rule fixed now, applied after Study 2).
Layouts: $5{,}100{,}000 + 1{,}000p + K$.
Analysis: Study-1 criteria per arm and density; plus a between-run summary over the 5 seeds (interquartile mean and a bootstrap over seeds), so variation across training runs is reported, not only layout sampling.

4. Study 4: communication and sensing

Conditions (V0 and the Study-3 arm, Study-1 actors): communication range 1.5 m (reference), 0.6 m and 0.45 m (the range rule requires ≥ 0.40 m); dropout probability 0.1 and 0.3 per pair and period (missing neighbour replaced by its last received state); one-period delay of neighbour states. Dropouts and delay violate A3, so no guarantee is claimed for them.
Primary endpoint: collision rate per condition with 95% Wilson interval (descriptive; the question is whether collisions reappear), plus GOAL and sustained nonprogress.
Layouts: $5{,}200{,}000 + 1{,}000p + K$; 500 per actor and density per condition.

5. Compute estimate

Study 1 ran 27,000 paired trials in three arms in about 13 minutes (about 50 s per 1,000 filter-on trials). Study 2: 7 arms $\times$ 27,000 trials $\approx$ 3 h, plus gain sweeps $\approx$ 1 h. Study 3: training $\approx$ 2.7 h per seed (Study-1 seed 0) $\times$ 5 seeds $\approx$ 14 h, evaluation $\approx$ 1 h. Study 4: 5 conditions $\times$ 2 arms $\times$ 13,500 trials $\approx$ 2 h. Code and tests for the variants: about 1–2 weeks.

6. Code to write and verify before locking

- One-sided margin (V1) and ZR rows (V2), with unit tests that the zero command is admissible where claimed and a written proof for V2 for advisor check.
- Fallback and recovery rule (V3); filter-only ℓ separated from the layout sampler and environment fingerprint (V4), with a test that layouts are byte-identical to the Study-1 sampler.
- Range, dropout and delay models (Study 4); confirm the Study-1 configuration reproduces the Study-1 trial records exactly when all new options are off.

7. Study-1 findings that informed this design (to be disclosed)

Lost completions were mostly intervention-active nonprogress near obstacles; every infeasible QP was a three-row conflict at the stand-off; the methodology report estimated the obstacle margin at about $13\times$ a tight one-sided margin; the actors failed the competence gate; neighbour pairs were almost always within the 1.5 m range.

8. Decisions requested from the advisor

1. Keep the Study-1 margins (5 pp GOAL, 0.01 m formation) or set new ones, with a rationale.
2. Approve the arms V1–V5, the recovery rule and the Study-3 selection rule; add or drop arms.
3. Competence gate threshold and training budget for Study 3; whether to add a filter-aware training arm.
4. Whether Study 4 is in scope, and whether $N = 5$ robots should be added.